

PROGRAMMING FUNDAMENTALS

Spring 2024

20th March

An Array

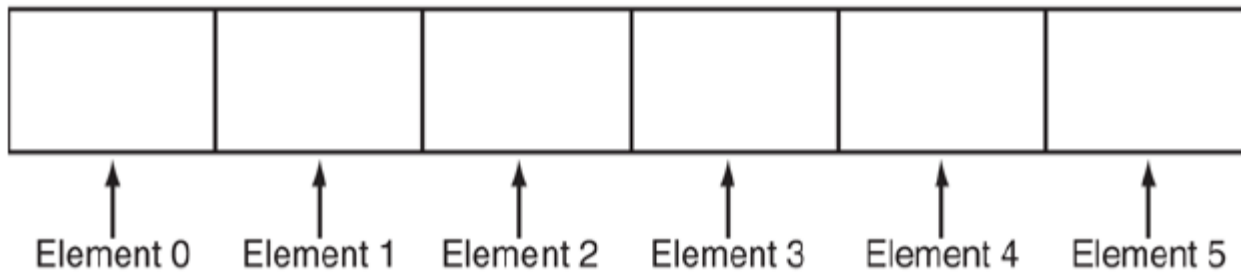
- An array allows you to store and work with multiple values of the **same** data type
- The variables discussed earlier are designed to hold only one value at a time
- An array works like a variable that can store a group of values, all of the same type
- The values are stored together in consecutive memory locations

Declaring an array size

```
int days[6];
```

- The name of this array is days
- The number inside the brackets is the array's *size declarator*
- It indicates the number of *elements*, or values, the array can hold
- The days array can store six elements, all of type integer

Declaring an array



Declaring an array size

- An array's size declarator must be a constant integer expression with a value greater than zero
- It can be either a literal, as in the previous example, or a named constant

Declaring an array size

```
const int Arr_size = 10;  
int num[Arr_size];
```

```
int arraySize;
```

```
cout << "Enter the size of the array: ";  
cin >> arraySize;  
cout << endl;
```

```
int list[arraySize];
```

Declaring Arrays

- Arrays of any data type can be defined:

```
float temperatures[100]; // Array of 100 floats
```

```
string names[10]; // Array of 10 string objects
```

```
long units[50]; // Array of 50 long integers
```

```
double sizes[1200]; // Array of 1200 doubles
```

Array Elements

- The individual elements of an array are assigned unique subscripts
- These subscripts are used to access the elements
- A subscript is used as an index to pinpoint a specific element within an array
- The first element is assigned the subscript 0, the second element is assigned 1, and so forth. The six elements in the array
- hours would have the subscripts 0 through 5.

Arrays

- The expression `hours[0]` is pronounced “hours sub zero.”

`hours[0] = 20;`

- “hours sub zero is assigned twenty.”

`hours[3] = 30;`

- ???

Initializing an array

```
for (int index = 0; index < 10; index++)  
{  
    sales[index] = 0.0;  
}
```

Reading data into an array

```
for (int index = 0; index < 10; index++)  
{  
    cin >> sales[index];  
}
```

Printing an array

```
for (int i = 0; i < 10; i++)  
{  
    cout << sales[index] <<endl;  
}
```

Finding the sum and average of an array

```
sum = 0;  
for (int index = 0; index < 10; index++)  
    sum = sum + sales[index];  
average = sum / 10;  
cout<<" Average = "<<average;
```

```
#include <iostream>

using namespace std;
int main()
{
    int sales[10];
    int i;
    float sum = 0.0;
    float average = 0.0;
    sum = 0;
    for (i = 0; i < 10; i++)
        sales[i] = i;

    for (i=0; i<10; i++)
        sum = sum + sales[i];
    average = sum / 10;
    cout<<"Average = "<<average;
    return 0;
}
```

Array initialization during declaration

```
float sales2[5] = {1.2, 2.2, 3.2, 4.2, 5.2};
```

When initializing in declaration, it is not necessary to write the size of the array.

Size is determined by the number of initial values in the braces

But square brackets must be included following the array name

```
float sales2[] = {1.2, 2.2, 3.2, 4.2, 5.2};
```

Array initialization during declaration

```
int num[10] = {0};
```

Declares num to be an integer array of size 10 components and initializes all components to zero

Example

Lec-9.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      cout<<"The initial Contents of the Array "<<endl;
7      int MyArray[5] = {0};
8      for (int i = 0; i<5; i++)
9          cout<<MyArray[i]<<endl;
10     return 0;
11 }
```

The initial Contents of the Array

0

0

0

0

0

:

Partial Initialization of arrays

```
int num[10] = {2, 3, 4};
```

Declares num to be an integer array of size 10 components and initializes num[0] to 2, num[1] to 3, num[2] to 4 and all other components to 0

```
int num[] = {2, 3, 4};
```

Declares num to be an integer array of size 3 and initializes num[0] to 2, num[1] to 3, num[2] to 4

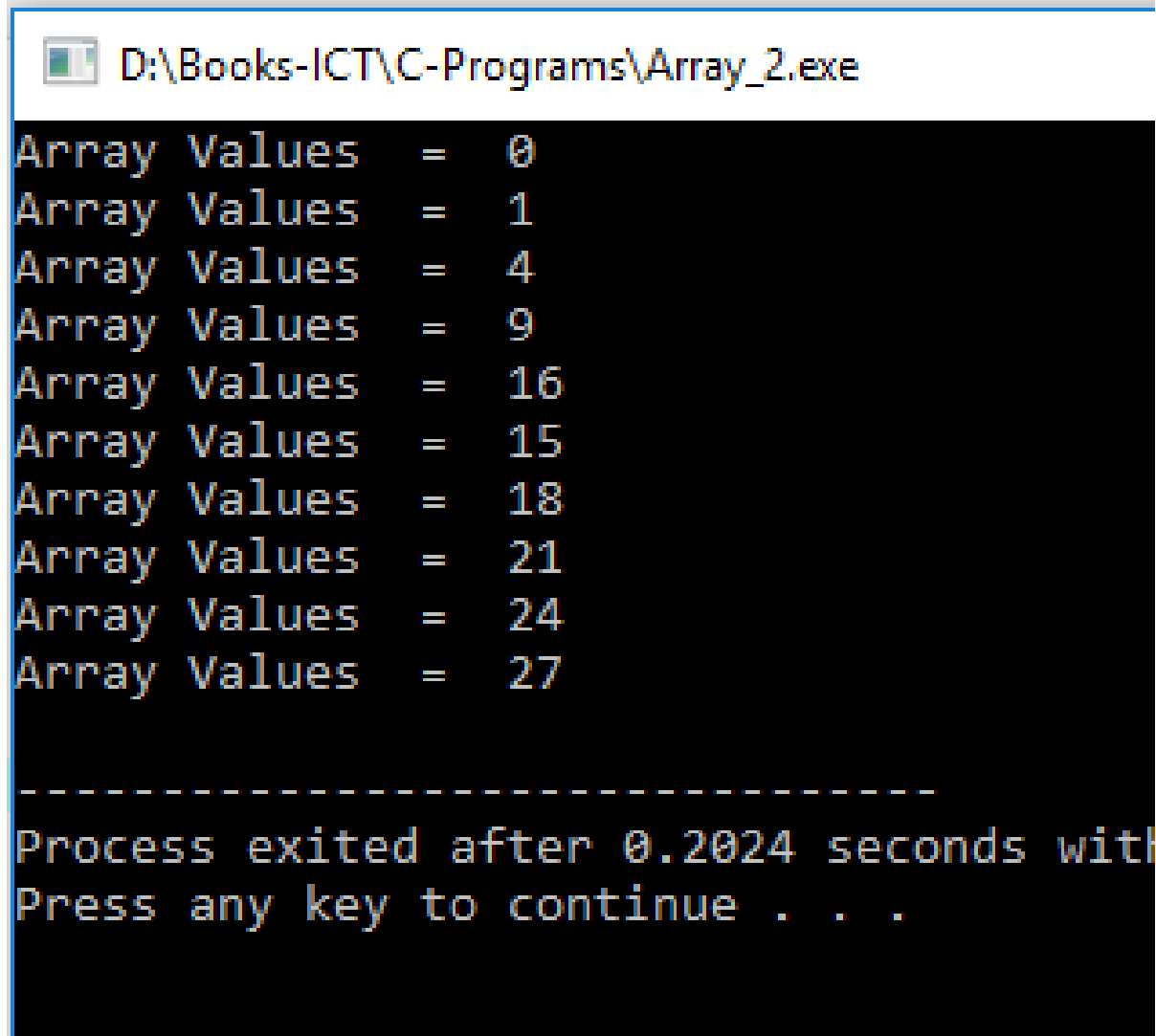
```
int num[12] = {2, 3, 4};
```

Declares num to be an integer array of size 12 and initializes num[0] to 2, num[1] to 3, num[2] to 4 and remaining components/cells in array are initialized to zero

Exercise#1

- Write a program that declares an array AAA of 10 components of type **double**
- Initialize the array so that the first 5 components are equal to the square of the index variable, and the last 5 components are equal to three times the index variable. Output the array so that one elements per line is printed

Sample Run..



```
D:\Books-ICT\C-Programs\Array_2.exe  
Array Values = 0  
Array Values = 1  
Array Values = 4  
Array Values = 9  
Array Values = 16  
Array Values = 15  
Array Values = 18  
Array Values = 21  
Array Values = 24  
Array Values = 27  
  
-----  
Process exited after 0.2024 seconds with  
Press any key to continue . . .
```

Book Reference

- D.S. Malik (ch - 4)
- Tony Gaddis (ch - 4)